CoalVision India:

Coal Stock Analytics for Thermal Power Plants

Project Overview

Coal remains the primary fuel source for India's thermal power generation. Maintaining adequate coal stock is essential to ensure uninterrupted electricity production and energy security. However, power plants frequently experience stock shortages, transport delays, and supply-demand mismatches.

This project analyses coal stock data from thermal power stations across India between 2018 and 2025. Using Python for data processing and Power BI for visualization, the dashboard identifies stock adequacy trends, sector-wise performance, transport dependencies, and regional stress zones to support data-driven decision-making.

Business Problem

Thermal power plants require sufficient coal inventory to meet daily generation requirements. Insufficient stock levels can lead to:

- Reduced power generation
- Increased operational risk
- Supply chain disruptions
- Higher dependence on emergency coal sourcing

The objective of this project is to assess whether coal availability is keeping pace with power generation capacity and identify the factors contributing to stock shortages.

Project Objectives

1. To analyse the relationship between installed capacity, coal requirement, and consumption.
2. To evaluate stock adequacy across thermal power plants.
3. To identify sectors and states facing critical coal shortages.
4. To understand transportation dependencies affecting coal movement.
5. To compare actual coal stock against normative requirements.
6. To generate actionable recommendations for improving coal supply reliability.

Data Preparation & Methodology

The analysis was performed using:

Python

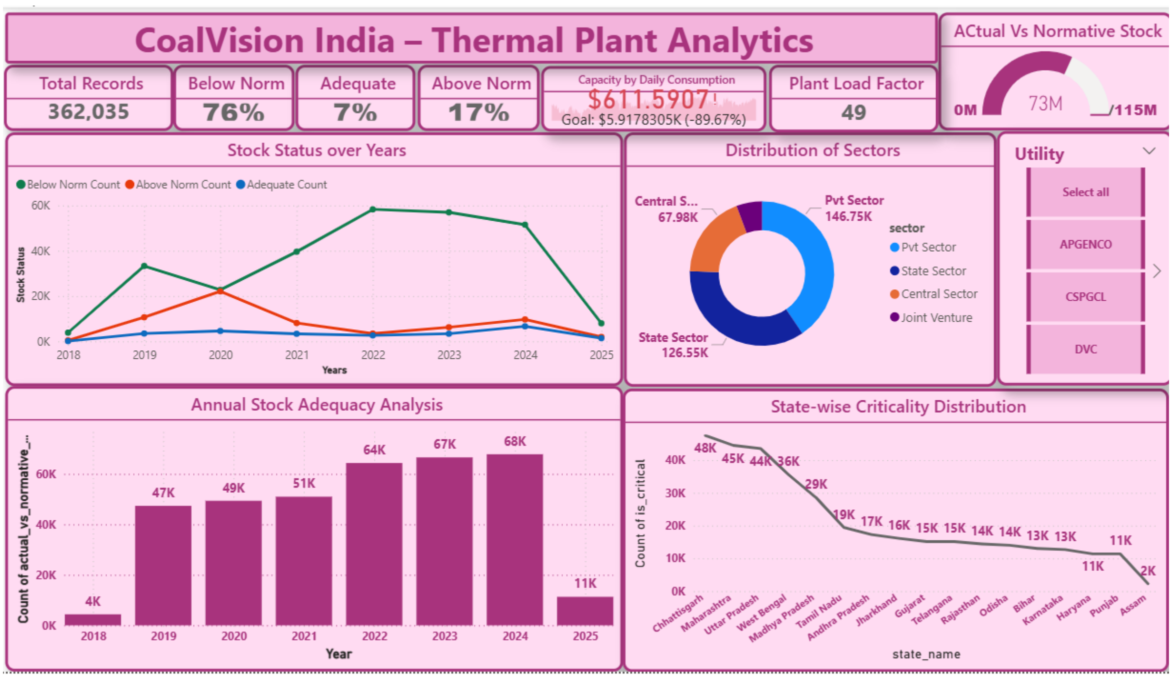
- Data Cleaning
- Missing Value Treatment
- Feature Engineering
- Exploratory Data Analysis
- Statistical Aggregation

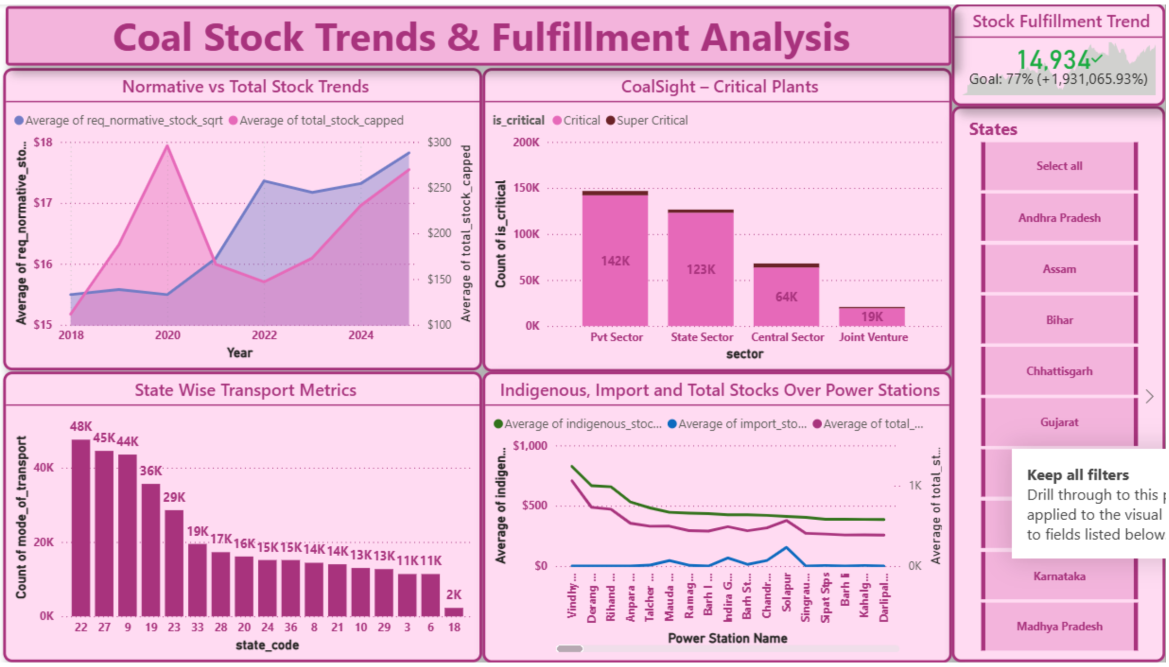
Power BI

- KPI Dashboards
- Trend Analysis
- Sector Comparisons
- Geographic Insights
- Interactive Filtering

The final dashboard integrates Python-generated analytics into Power BI to provide an interactive and comprehensive decision-support system.

Insights from Dashboard





Chapter 1: Capacity Drives Coal Demand

- The dashboard begins with three key performance indicators:
- Average Installed Capacity: ~1168 MW
- Average Daily Coal Requirement: ~15 Tons
- Average Daily Coal Consumption: ~3 Tons
- The analysis reveals a strong relationship between plant capacity and coal demand.

Insight

- As plant capacity increases, coal requirement increases proportionally. However, actual consumption remains below expected levels.

Interpretation

- This indicates either,
  - Under-utilization of generating assets or
  - Coal supply constraints limiting operations.

Chapter 2: The Stock Adequacy Challenge

- Stock adequacy was analysed by comparing actual stock against normative requirements.

Findings

- Approximately 76% of plants operate below normative stock levels.

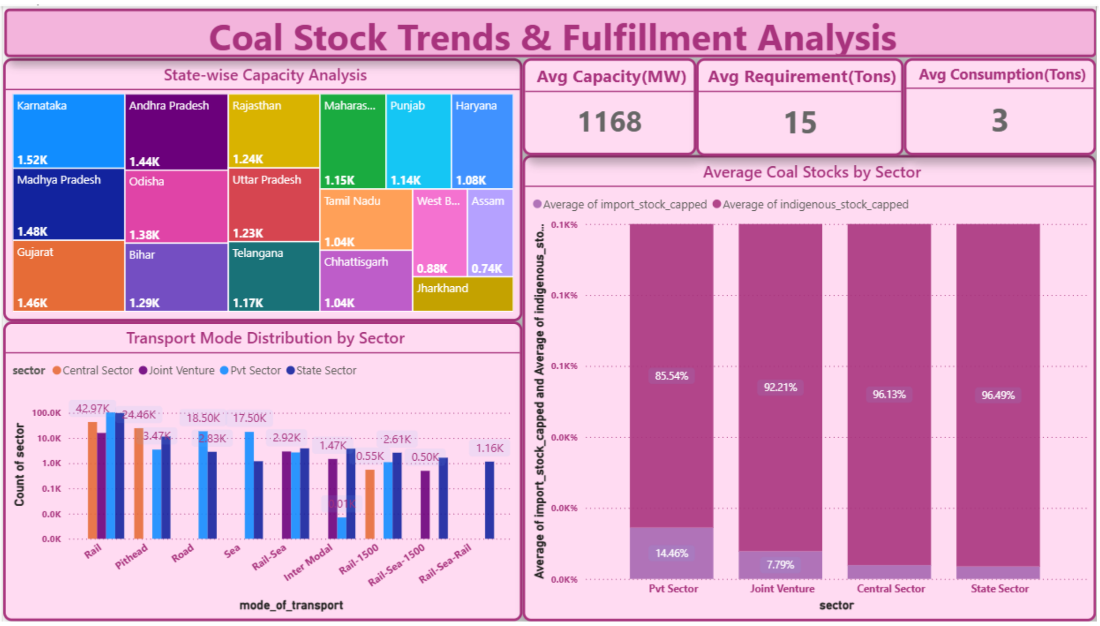
• Only a small percentage maintain adequate or above-normal stock.

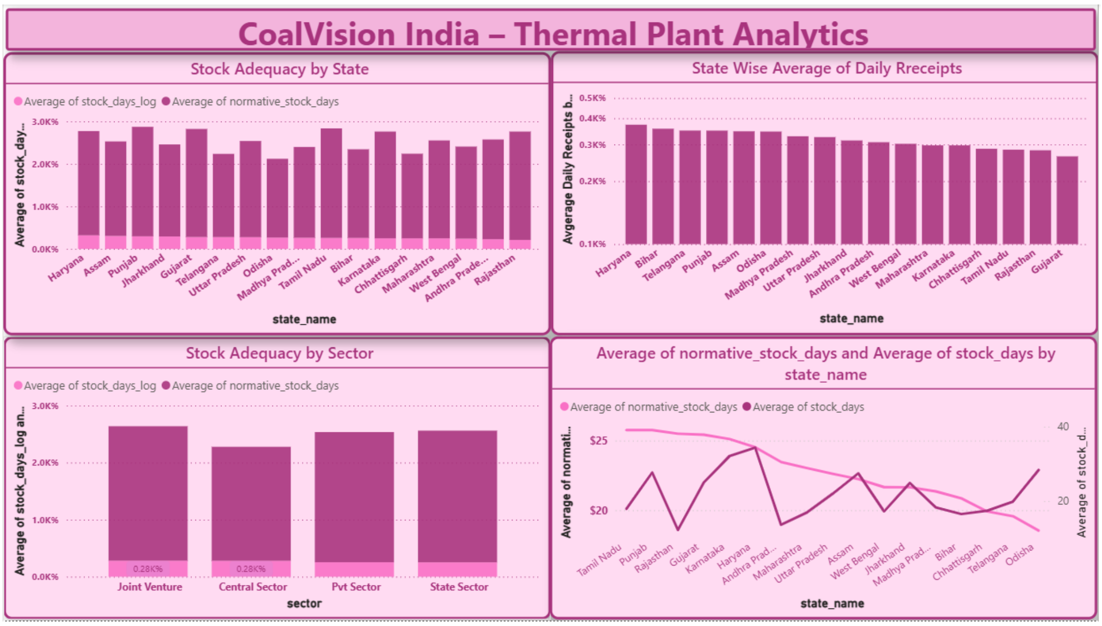
Insight

• Coal shortages are not isolated incidents but a recurring pattern across multiple years.

Business Impact

• Persistent stock deficits increase the risk of power generation interruptions and reduce operational resilience.





Chapter 3: Sector-Wise Performance

The analysis compared coal stock conditions across sectors:

- State Sector
- Central Sector
- Private Sector
- Joint Venture Sector

Findings

- State sector plants maintain relatively balanced stock levels.
- Private and Central sector plants show higher proportions of critical stock conditions.

Insight

- Operational practices and supply chain effectiveness vary significantly across sectors.

Business Impact

Targeted intervention is required in sectors experiencing repeated shortages.

Chapter 4: Regional Stress Zones

- State-wise analysis revealed geographical concentration of stock shortages.

High-Risk States

- Chhattisgarh
- Maharashtra
- Uttar Pradesh

- These states consistently report higher numbers of critically stocked plants.

Insight

- Many of these states also have significant thermal generation capacity, making shortages particularly impactful.

Business Impact

- Regional coal supply planning should prioritize these high-risk zones.

Chapter 5: Transportation Dependency

- Coal transportation plays a critical role in maintaining stock adequacy.

Findings

Transport Mode Distribution:

- Rail: Dominant mode (>60%)
- Road: Secondary mode
- Pithead: Limited usage

Insight

The coal ecosystem is heavily dependent on rail infrastructure.

Business Impact

- Any disruption in rail logistics can rapidly create stock shortages across multiple power plants.

Chapter 6: Normative Stock vs Actual Stock

- A year-wise comparison was performed between actual coal stock and normative requirements.

Findings

- Normative stock consistently exceeds actual stock.
- Gap remains present throughout the study period.
- Slight improvement observed after 2023.

Insight

- Supply efficiency has improved marginally but remains insufficient to fully meet operational requirements.

Business Impact

- Long-term supply planning remains necessary to bridge the stock adequacy gap.

Key Insights

- Capacity growth is increasing coal demand.
- Approximately 76% of plants operate below normative stock levels.
- Private and Central sectors experience greater stock criticality.
- Rail transport dominates coal movement, creating dependency risks.
- Chhattisgarh, Maharashtra, and Uttar Pradesh are key stress regions.

- Actual stock consistently falls below normative requirements.
- Post-2023 trends indicate gradual improvements in supply efficiency.

Recommendations

Logistics Optimization

- Improve rail scheduling, rake allocation, and multimodal transportation strategies.

Indigenous Coal Sourcing

- Increase domestic coal production and reduce dependency on external supply sources.

Dynamic Stock Monitoring

- Implement real-time monitoring systems to identify shortages before they become critical.

Predictive Analytics

- Use forecasting models to predict future demand-supply mismatches and proactively manage inventory.

Automated Reporting

- Integrate automated analytics pipelines for faster stakeholder reporting and decision-making.

Business Value Delivered

The CoalVision India dashboard provides:

- Real-time visibility into coal stock health
- Early identification of critical shortage zones
- Sector-wise performance benchmarking
- Transport dependency analysis
- Data-driven recommendations for improving energy security

By combining Python analytics with Power BI interactivity, the solution transforms raw coal stock data into actionable intelligence for policymakers, utility operators, and energy planners.

Final Conclusion

- India's thermal power ecosystem continues to expand, but coal supply stability has not kept pace with capacity growth.
- The analysis highlights persistent stock shortages, strong dependence on rail logistics, and regional concentration of critical stock conditions. While recent years show signs of improvement, significant gaps remain between actual and normative stock levels.
- Ensuring reliable power generation will require strategic investments in logistics optimization, domestic coal sourcing, predictive planning, and real-time stock monitoring.
- The CoalVision India Dashboard serves as a decision-support platform that enables stakeholders to proactively manage coal inventories, reduce operational risks, and strengthen India's energy security.